

MOHAMED H. ALY

EXPERIENCE

E& (ETISALAT UAE)

Senior Software Developer

Dec 2025 - Present

- **PORTAL MODERNIZATION:** Led migration of a 15-year-old Java Server Faces CIM portal to React with micro-frontend architecture, each module backed by a dedicated BFF microservice.
- **COMPONENT LIBRARY OVERHAUL:** Shrunk a bloated internal component library from 50 MB to under 5 MB by eliminating redundancy and dead code, preserving 100% of original functionality.
- **ALGORITHM OPTIMIZATION:** Optimized cell selection logic from $O(n)$ to $O(1)$, replacing linear scans with direct index lookups in interactive data grids.
- **TECHNICAL LEADERSHIP:** Drove React adoption within a team transitioning from a legacy JSF codebase, establishing patterns and standards that accelerated delivery across all micro-frontend modules.

THE ADDRESS HOLDING

Frontend Engineer

Nov 2024 – Dec 2025

- **Internal system development:** Built and maintained CRMs and HRMs using Next.js, focusing on performance, scalability, and maintainability.
- **Project ownership:** Spearheaded the development of a property listing e-commerce platform, authoring over 85% of the production code.
- **SEO optimization:** Implemented server-side rendering and dynamic routing to maximize organic visibility and align with SEO best practices.
- **Adaptability:** Legacy modernization: Maintained and extended a legacy Vue.js codebase, refactoring outdated patterns and ensuring seamless integration with modern backend services.

SIDEUP

Senior Frontend Engineer

Dec 2023 – Nov 2024

- **Mentorship & leadership:** Guided junior engineers, cultivating a collaborative environment and improving team efficiency.
- **Hands-on coding leadership:** Distributed tasks effectively and tackled complex features to ensure robust, maintainable code.
- **Stakeholder collaboration:** Aligned dashboard development with user needs and business objectives through proactive communication.
- **Process optimization:** Introduced workflow improvements to enhance productivity, efficiency, and code reliability.

SIDEUP

Frontend Engineer

Feb 2023 – Dec 2023

- **B2C dashboard development:** Led frontend implementation of a next-generation shipping dashboard using React, TypeScript, and Bootstrap.
- **UX/UI excellence:** Partnered with designers to deliver an intuitive interface, improving usability and customer satisfaction.
- **Quality assurance:** Collaborated with QA teams to conduct thorough testing and debugging for optimal performance.
- **Adaptability:** Quickly mastered Next.js and Material UI to modernize legacy features and deliver consistent improvements.

ADVANCED COMPUTER TECHNOLOGIES (ACT)

Interface Specialist

Feb 2022 – Feb 2023

- **System integration:** Integrated hotel subsystems (Door Locks, IPTV, POS) with PMS solutions to streamline operations for global clients.
- **Agile collaboration:** Worked within agile teams to prioritize stakeholder requirements and ensure smooth delivery cycles.
- **Remote troubleshooting:** Diagnosed and resolved integration issues via log and error analysis, minimizing downtime.
- **Project coordination:** Led the integration of Oracle's Opera Web Services across 480+ IHG hotels globally within four months, coordinating complex technical dependencies and stakeholders.

SELF-EMPLOYED

Freelance Frontend Engineer

2021 – Present

- **Payment Dashboard:** Aman Accept Dashboard (Next.js) — a payment operations and analytics tool.
- **Corporate Website:** Beltone Company Website (Next.js) — SEO-optimized and performance-driven.
- **Blockchain E-commerce:** D'art Space (Angular + crypto) — a blockchain-based art e-commerce platform.
- **CRM Platform:** Built a full-featured CRM using Vue.js, delivering a scalable and maintainable system with a focus on performance and user experience.
- **Timely delivery:** delivered client projects using Next.js, Vue, Angular, and TypeScript, with focus on scalability and SEO.
- **Team Collaboration:** Worked closely with designers and backend teams to deliver pixel-perfect, responsive UIs.
- **Client Relations:** Maintained ongoing client partnerships through reliable delivery and technical guidance.

TECHNICAL SKILLS

FRONTEND

HTML5, CSS3, SASS, Bootstrap, Tailwind, JavaScript, TypeScript, React, Redux, React Query, Vue, Angular, Next.js, Three.js, WebGL, Framer Motion, GSAP, Responsive Design (Flexbox, Grid, Media Queries), Accessibility (a11y) & WCAG compliance, Performance Optimization (Lighthouse, Core Web Vitals), Server-Side Rendering (SSR) & Static Site Generation (SSG), Internationalization (i18n), Progressive Web Apps (PWA), Authentication/Authorization (JWT, OAuth2), RESTful & GraphQL API Design Principles, Storybook, Webpack, Vite, SWC, Babel

BACKEND

Node.js, Express, Nest, PHP, C, RESTful APIs, WebSockets / Real-time APIs, Clean Architecture & SOLID Principles

DATABASES & DEVOPS

MongoDB, MySQL, PostgreSQL, Redis, Firebase, Prisma, TypeORM, Docker, Docker Compose, Serverless, Bash CLI, GitHub Actions, CI/CD Integration (Vercel, Netlify, Cloudflare Pages)

TOOLS & CONCEPTS

Git & GitHub, Jira, Agile / Scrum, Unit Testing, Design Systems (Figma → Code handoff, Storybook integration), Version Control Workflows (Git Flow, trunk-based development), Type Safety & Code Quality (ESLint, Prettier, Husky), Performance Profiling Tools (Chrome DevTools, React Profiler, Angular DevTools), Data Structures & Algorithms, AI & APIs (OpenAI / LLM integrations)

EDUCATION

BACHELOR OF SCIENCE (B.S), MANAGEMENT INFORMATION SYSTEMS, GPA 3.1

July 2021

Arab Academy for Science & Technology & Maritime Transport (AASTMT)

PROJECTS & COURSES/CERTIFICATES

Projects and certifications available in my [online portfolio](#).